MAST20030 Differential Equations
— Problem Sheet 3: Laplace Transforms —

1. Determine whether the following integrals converge or diverge:

   (a) $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}t}{(1+t^2)^5}$ (b) $\displaystyle\int_{1}^{\infty} \frac{\mathrm{e}^t}{t^7}\,\mathrm{d}t$

   (c) $\displaystyle\int_{0}^{\infty} \mathrm{e}^{-t}\sin t\,\mathrm{d}t$ (d) $\displaystyle\int_{0}^{\infty} t\mathrm{e}^{-t}\log t\,\mathrm{d}t$

2. Compute the Laplace transforms of the following functions and indicate the domain of the transformed function:

   (a) e^{-5t} (b) $\cos(2t)$ (c) $-3\sin(4t)+7\mathrm{e}^{3t}$

   (d) $\mathrm{e}^t\sin t$ (e) $2\sinh(3t)\sin(3t)$

3. Determine the inverse Laplace transform of the following functions, all assumed to be defined for $\Re(s)$ sufficiently large:

   (a) $\dfrac{1}{(s-2)(s^2+1)}$ (b) $\dfrac{1}{s^2(s^2+1)}$ (c) $\dfrac{1}{s^3-8}$

   (d) $\dfrac{3s-8}{s^2-5s+6}$ (e) $\dfrac{s^2+20s+9}{(s-1)^2(s^2+9)}$ (f) $\dfrac{3s-2}{s^2-2s+10}$

4. Using the integral definition, find the Laplace transforms of

   (a) $\begin{cases} kt/c, & 0 \le t \le c \\ 0, & t > c \end{cases}$ (b) $\begin{cases} k, & 0 < a \le t \le b \\ 0, & \text{elsewhere} \end{cases}$

   (c) $\sinh(at)$ (d) $\cos(at)$

5. Use the tables to find the Laplace transforms of

   (a) $2t^2+6-3e^{4t}$ (b) $A\sin(\omega t)+B\cosh(kt)$

   (c) t^4+t^5 (d) $\cos^2 t$

6. Use the tables to find the inverse Laplace transforms of

   (a) $\dfrac{3}{s^2}$ (b) $\dfrac{4}{s^2+16}$

   (c) $\dfrac{s}{s^2+9}$ (d) $\dfrac{5s}{s^2-25}$

7. Use partial fractions to find the inverse Laplace transforms of

   (a) $\dfrac{s}{s^2+3s-4}$ (b) $\dfrac{1}{(s-2)(s+1)}$

8. Using the s-shifting theorem, find the Laplace transforms of

   (a) $3t\mathrm{e}^{2t}$ (b) $5e^{-6t}\sin(4t)$

   (c) $2e^{-t}\sinh(3t)$ (d) $2\sinh t\cos t$

9. Use the s-shifting theorem to find the inverse Laplace transforms of

   (a) $\dfrac{3}{(s+2)^2}$ (b) $\dfrac{1}{(s-2)^3}$

   (c) $\dfrac{1}{(s-4)^2+1}$ (d) $\dfrac{s+2}{(s+2)^2+4}$

10. Using the t-shifting theorem, find the Laplace transforms of

    (a) $\cosh(t-1)H_1(t)$ (b) $tH_\pi(t)$

11. Use the t-shifting theorem to find the inverse Laplace transforms of

    (a) $\dfrac{\mathrm{e}^{-3s}}{s}$ (b) $\left(\dfrac{5s+4}{s^2}\right)\mathrm{e}^{-2s}$

12. Rewrite the following functions in piecewise-defined form (i.e., don't use Heavisides):

    (a) $H(t)+3H_1(t)-H_2(t)$ (b) $t^2H_2(t)$ (c) $\sin(2t-\frac{2\pi}{3})H_{\pi/3}(t)$

    (d) $(t-5)^2(1-2H_4(t))$ (e) $\sin(H(t))$ (f) $H(\sin(t))$

13. Compute the Laplace transforms of the following piecewise-defined functions and indicate the domain of the transformed function:

    (a) $\begin{cases} t & \text{if } 0\le t\le 1,\\ 1 & \text{if } t>1,\end{cases}$ (b) $\begin{cases} \sin t & \text{if } 0\le t\le \pi,\\ 0 & \text{if } t>\pi,\end{cases}$

    (c) $\begin{cases} t & \text{if } 0\le t\le 1,\\ 2-t & \text{if } 1<t\le 2,\\ 0 & \text{if } t>2,\end{cases}$ (d) $\begin{cases} 1 & \text{if } 0\le t\le 2,\\ 0 & \text{if } 2<t<3,\\ -1 & \text{if } 3\le t\le 6,\\ 0 & \text{if } t>6.\end{cases}$

14. Use Laplace transforms to solve the following IVPs:

    (a) $y''(t)+2y'(t)-3y(t)=32\mathrm{e}^{5t}$ where $y(0)=0$ and $y'(0)=12$

(b) $y''(t) + 9y(t) = 6\cos(3t)$ where $y(0) = 1$ and $y'(0) = 6$

(c) $y''(t) - 2y'(t) + 2y(t) = 2t - 1$ where $y(0) = \dfrac{1}{2}$ and $y'(0) = 3$

(d) $y''(t) + 4y'(t) + 5y(t) = 25t$ where $y(0) = -5$ and $y'(0) = 7$

(e) $y'''(t) - y''(t) + 9y'(t) - 9y(t) = 0$ where $y(0) = 0$, $y'(0) = 3$ and $y''(0) = 6$

(f) $y''''(t) - 2y''(t) + y(t) = 8\mathrm{e}^t$ where $y(0) = 1$, $y'(0) = 2$, $y''(0) = 1$ and $y'''(0) = 10$

15. Use Laplace transforms to solve the following IVPs where $f(t) = \begin{cases} 1 & \text{if } t < 1, \\ 0 & \text{if } t > 1 \end{cases}$:

(a) $y'(t) + y(t) = f(t)$ where $y(0) = 0$

(b) $y''(t) + 4y(t) = (1 - H_{2\pi}(t))\cos t$ where $y(0) = 0$, and $y'(0) = 0$

(c) $y''(t) + 4y(t) = f(t)$ where $y(0) = 1$ and $y'(0) = 1$

(d) $4y''(t) + 9y(t) = 5\delta_2(t)$ where $y(0) = 2$ and $y'(0) = 0$

(e) $y''(t) + 6y'(t) + 9y(t) = 2\mathrm{e}^{-3t} + 4\delta_2(t) + 4\delta_4(t) + 4\delta_6(t)$ where $y(0) = 0$ and $y'(0) = 0$

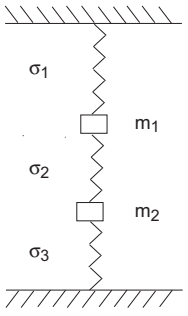
16. Use Laplace transforms to solve the following IVPs where $g(t)$ is left undefined:

(a) $y''(t) + \omega^2 y(t) = g(t)$ where $y(0) = 1$ and $y'(0) = 0$

(b) $y''(t) + 3y'(t) + 2y(t) = g(t)$ where $y(0) = 0$ and $y'(0) = 0$

(c) $y''(t) + 2y'(t) + y(t) = g(t)$ where $y(0) = 1$ and $y'(0) = 1$

17. A mechanical system consists of three springs, σ_1, σ_2 and σ_3 of negligible mass, equal natural length and spring constant 1N/m, and two objects, m_1 and m_2 of mass 1 kg each.



Assuming that the damping is negligible, and that the departure of m_1 from its equilibrium position is $y_1(t)$, and that of m_2 is $y_2(t)$ at time t. The equations of motion are given by the following system of second order ordinary differential equations:

$$\begin{aligned} y_1'' &= y_2 - 2y_1 \\ y_2'' &= y_1 - 2y_2 \end{aligned}$$

Assume the initial conditions are

$$\begin{aligned} y_1(0) &= 1, \qquad y_1'(0) = \sqrt{3} \\ y_2(0) &= 1, \qquad y_2'(0) = -\sqrt{3} \end{aligned}$$

Use Laplace transforms to solve the system of differential equations subject to the given initial conditions.

18. An object with mass m receives 11 impulses of strength p at 1 second intervals at $t = 0$, $t = 1$, $t = 2$, ..., $t = 10$. The differential equation describing the motion of this object is

$$m\frac{dv}{dt} = p\sum_{k=0}^{10}\delta(t-k)$$

If the object is initially at rest, find its velocity at time $t \geq 0$.

19. Use the convolution theorem to find the Laplace transforms of

(a) $f(t) = \int_0^t e^{\tau}\,(t-\tau)\,d\tau$

(b) $f(t) = \int_0^t \sin(t-\tau)e^{3\tau}\,d\tau$

(c) $f(t) = \int_0^t \delta(t-\tau)\sinh(2\tau)\,d\tau$

(d) $f(t) = \int_0^t (1+\tau)(t-\tau)^2\,d\tau$

20. Solve the integral-differential equation

$$\int_0^t y(\tau)\,d\tau - y'(t) = t$$

for $t \geq 0$, with $y(0) = 4$.

21. In a series RLC-circuit with resistance $R = 2\Omega$, inductance $L = 1H$, capacitance $C = 0.25F$ and voltage $v = 50V$, the current i at time t is determined by the integral-differential equation

$$2i + \frac{di}{dt} + 4\int_0^t i(\tau)\,d\tau = 50$$

Solve the integral-differential equation to find the current, if initially there is no current in the circuit.

22. Consider an RC circuit, with resistance R and capacitance C. The voltage v at time t is always zero, except from $t = a$ to $t = b$ when $v = v_0$ is a constant. The current i at any time is given by the differential equation

$$Ri + \frac{q}{C} = v$$

where q is the charge on the capacitor at any time and

$$v(t) = v_0[H_a(t) - H_b(t)]$$

Since $i = \dfrac{dq}{dt}$, the differential equation becomes

$$Ri + \frac{1}{C}\int_0^t i(\tau)\,d\tau = v_0[H_a(t) - H_b(t)]$$

Solve the integral equation for the current.

MAST20030 Differential Equations
— Solutions to Problem Sheet 3: Laplace Transforms —

1.

(a) Converge (b) Diverge

(c) Converge (d) Converge

2.

(a) $\dfrac{1}{5+s}$, $s > -5$ (b) $\dfrac{s}{s^2+4}$, $s > 0$ (c) $\dfrac{7}{s-3} - \dfrac{12}{16+s^2}$, $s > 3$

(d) $\dfrac{1}{1+(s-1)^2}$, $s > 1$ (e) $\dfrac{36s}{324+s^4}$

3.

(a) $\dfrac{1}{5}(\mathrm{e}^{2t} - \cos t - 2\sin t)$ (b) $t - \sin t$

(c) $\dfrac{1}{12}\mathrm{e}^{-t}(\mathrm{e}^{3t} - \cos\sqrt{3}t - \sqrt{3}\sin\sqrt{3}t)$ (d) $2\mathrm{e}^{2t} + \mathrm{e}^{3t}$

(e) $\dfrac{1}{5}(\mathrm{e}^{t}(8+15t) - 8\cos 3t - 6\sin 3t)$ (f) $\dfrac{1}{3}\mathrm{e}^{t}(9\cos 3t + \sin 3t)$

4.

(a) $\dfrac{k(1-\mathrm{e}^{-cs}(1+cs))}{cs^2}$, $s > 0$ (b) $\dfrac{k(\mathrm{e}^{-as} - \mathrm{e}^{-bs})}{s}$, $s > 0$

(c) $\dfrac{a}{s^2-a^2}$, $s > a$ (d) $\dfrac{s}{s^2+a^2}$, $s > 0$

5.

(a) $\dfrac{6}{s} + \dfrac{4}{s^3} - \dfrac{3}{s-4}$, $s > 4$ (b) $\dfrac{Bs}{s^2-k^2} + \dfrac{A\omega}{s^2+\omega^2}$, $s > k$

(c) $\dfrac{120}{s^6} + \dfrac{24}{s^5}$, $s > 0$ (d) $\dfrac{2+s^2}{4s+s^3}$, $s > 0$

6.

(a) $3t$ (b) $\sin 4t$

(c) $\cos 3t$ (d) $5\cosh 5t$

7.

(a) $\dfrac{4\mathrm{e}^{-4t}}{5} + \dfrac{\mathrm{e}^{t}}{5}$

(b) $\dfrac{\mathrm{e}^{2t}}{3} - \dfrac{\mathrm{e}^{-t}}{3}$

8.

(a) $\dfrac{3}{(s-2)^2}$, $s > 2$

(b) $\dfrac{20}{16 + (s+6)^2}$, $s > -6$

(c) $\dfrac{6}{s^2 + 2s - 8}$, $s > 2$

(d) $\dfrac{2(s^2-2)}{s^4+4}$, $s > 1$

9.

(a) $3\mathrm{e}^{-2t}t$

(b) $\dfrac{1}{2}\mathrm{e}^{2t}t^2$

(c) $\mathrm{e}^{4t}\sin t$

(d) $\mathrm{e}^{-2t}\cos 2t$

10.

(a) $\dfrac{\mathrm{e}^{-s}s}{s^2-1}$, $s > 1$

(b) $\dfrac{\mathrm{e}^{-\pi s}(1+\pi s)}{s^2}$, $s > 0$

11.

(a) $H_3(t)$

(b) $(4t-3)H_2(t)$

12.

(a) $\begin{cases} 0, & t < 0 \\ 1, & 0 \le t < 1 \\ 4, & 1 \le t < 2 \\ 3, & 2 \le t \end{cases}$

(b) $\begin{cases} 0, & t < 2 \\ t^2, & 2 \le t \end{cases}$

(c) $\begin{cases} 0, & t \le \frac{\pi}{3} \\ \sin\left(2t - \frac{2\pi}{3}\right), & \frac{\pi}{3} < t \end{cases}$

(d) $\begin{cases} (t-5)^2, & t < 4 \\ -(t-5)^2, & 4 \le t \end{cases}$

(e) $\begin{cases} 0, & t < 0 \\ \sin 1, & 0 \le t \end{cases}$

(f) $\begin{cases} 1, & 2n\pi < t \le (2n+1)\pi \\ 0, & (2n+1)\pi \le t < (2n+2)\pi \end{cases}$, $n \in \mathbb{Z}$

13.

(a) $\dfrac{1-\mathrm{e}^{-s}}{s^2}$, $s > 0$

(b) $\dfrac{1+\mathrm{e}^{-\pi s}}{1+s^2}$, $s > 0$

(c) $\dfrac{1+\mathrm{e}^{-2s}-2\mathrm{e}^{-s}}{s^2}$, $s > 0$

(d) $\dfrac{1-\mathrm{e}^{-2s}-\mathrm{e}^{-3s}+\mathrm{e}^{-6s}}{s}$, $s > 0$

14.

(a) $y = -2\mathrm{e}^{-3t} + \mathrm{e}^{t} + \mathrm{e}^{5t}$

(b) $y = \cos 3t + (2+t)\sin 3t$

(c) $y = \dfrac{1}{2} + t + 2\mathrm{e}^{t}\sin t$

(d) $y = 5t - 4 - \mathrm{e}^{-2t}\cos t$

(e) $y = \dfrac{1}{5}(3\mathrm{e}^{t} - 3\cos 3t + 4\sin 3t)$

(f) $y = t\mathrm{e}^{-t} + \mathrm{e}^{t}(1+t^{2})$

15.

(a) $y = 1 - \mathrm{e}^{-t} + (\mathrm{e}^{1-t} - 1)H_1(t)$

(b) $y = \dfrac{1}{3}(\cos t - \cos 2t)(1 - H_{2\pi}(t))$

(c) $y = \dfrac{1}{4}(1 + 3\cos 2t + 2\sin 2t + (\cos(2-2t) - 1)H_1(t))$

(d) $y = 2\cos\dfrac{3t}{2} - \dfrac{5}{6}\sin\left(3 - \dfrac{3t}{2}\right)H_2(t)$

(e) $y = t^{2}\mathrm{e}^{-3t} + 4(t-2)\mathrm{e}^{6-3t}H_2(t) + 4(t-4)\mathrm{e}^{12-3t}H_4(t) + 4(t-6)\mathrm{e}^{18-3t}H_6(t)$

16.

(a) $y = \cos\omega t + \dfrac{1}{\omega}\displaystyle\int_0^t g(t-\tau)\sin\omega\tau\,\mathrm{d}\tau$

(b) $y = \displaystyle\int_0^t (\mathrm{e}^{-\tau} - \mathrm{e}^{-2\tau})g(t-\tau)\mathrm{d}\tau$

(c) $y = (2t+1)\mathrm{e}^{-t} + \displaystyle\int_0^t \tau\mathrm{e}^{-\tau}g(t-\tau)\mathrm{d}\tau$

17.

$$y_1(t) = \cos t + \sin\sqrt{3}t, \qquad y_2(t) = \cos t - \sin\sqrt{3}t$$

18.

$$v = \frac{p}{m}\sum_{k=0}^{10} H_k(t)$$

19. (a) $\dfrac{1}{s^{2}(s-1)}$, $s > 1$ (b) $\dfrac{1}{(s^{2}+1)(s-3)}$, $s > 3$

(c) $\dfrac{2}{s^{2}-4}$, $s > 2$ (d) $\dfrac{2(s+1)}{s^{5}}$, $s > 0$

20.

$$y = 1 + 3\cosh t$$

21.

$$i = \frac{50}{\sqrt{3}} \mathrm{e}^{-t} \sin \sqrt{3} t$$

22.

$$i = \frac{v_0}{R} \left(\mathrm{e}^{\frac{1}{CR}(a-t)} H_a(t) - \mathrm{e}^{\frac{1}{CR}(b-t)} H_b(t) \right)$$